NVENC Application Note

Release 13.1

NVIDIA Corporation

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Contents

Chapter 1. Introduction

NVIDIA GPUs - beginning with the Kepler generation - contain a hardware-based encoder (referred to as NVENC in this document) which provides fully accelerated hardware-based video encoding and is independent of graphics/CUDA cores. With end-to-end encoding offloaded to NVENC, the graphics/CUDA cores and the CPU cores are free for other operations. For example, in a game recording scenario, offloading the encoding to NVENC makes the graphics engine fully available for game rendering. In the video transcoding use-case, video encoding/decoding can happen on NVENC/NVDEC in parallel with other video post-/pre-processing on CUDA cores.

The hardware capabilities available in NVENC are exposed through APIs referred to as NVENCODE APIs in the document. This document provides information about the capabilities of the hardware encoder and features exposed through NVENCODE APIs.

Chapter 2. NVENC Capabilities

NVENC can encode H.264 8-bit and H.264 10-bit, HEVC 8-bit, HEVC 10-bit, AV1 8-bit and AV1 10-bit contents. This includes motion estimation and mode decision, motion compensation and residual coding, and entropy coding. It can also be used to generate motion vectors between two frames, which are useful for applications such as depth estimation, frame interpolation, encoding using other codecs not supported by NVENC, or hybrid encoding wherein motion estimation is performed by NVENC and the rest of the encoding is handled elsewhere in the system. These operations are hardware accelerated by a dedicated block on GPU silicon die. NVENCODE APIs provide the necessary knobs to utilize the hardware encoding capabilities.

The tables below summarize the capabilities of the NVENC hardware exposed through NVENCODE APIs.

NVENC H264 Hardware Capabilities

Table 1: NVENC H264 Hardware Capabilities

Hardware Features	Ampere and TU GPUs	Ada GPUs	Blackwell GPUs	Jetson Thor
Baseline main and high profile: Capability to encode YUV 420 sequence	Y	Y	Y	Y
High10 profile: Support for encoding 10-bit content.	N	N	Y	N
4:2:2 encoding	N	N	Y	N
4:4:4 encoding (only CAVLC)	Y	Y	Y	Y
Lossless encoding	Y	Y	Y	Y

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Hardware Features	Ampere and TU GPUs	Ada GPUs	Blackwell GPUs	Jetson Thor
Motion estimation (ME) only mode: Capability to provide macro-block level motion vectors and intra or inter modes.	Y	Y	Y	Y
Field encoding	N	N	Y	N
Weighted prediction	Y	Y	Y	Y
Encoding support ARGB content	Y	Y	Y	Y
Multiple reference frames : Capability to use different reference frames	Y	Y	Y	Y

- **Y**: Supported, **N**: Not supported

NVENC HEVC Hardware Capabilities

Table 2: NVENC HEVC Hardware Capabilities

Hardware Features	Ampere and TU GPUs	Ada GPUs	Blackwell GPUs	Jetson Thor
Main profile: Capability to encode YUV 4:2:0 sequence.	Y	Y	Y	Y
Lossless encoding	Y	Y	Y	Y
Main10 profile: Support for encoding 10-bit content.	Y	Y	Y	Y
4:2:2 encoding	N	N	Y	N
4:4:4 encoding	Y	Y	Y	Y

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Table 2 – continued from previous page

Hardware Features	Ampere and TU GPUs	Ada GPUs	Blackwell GPUs	Jetson Thor
Motion estimation (ME) only mode: Capability to provide CTB level motion vectors and intra/inter modes.	Y	Y	Y	Y
8K encoding: Support for encoding 8192 x 8192 Content.	Y	Y	Y	Y
Weighted prediction	Y	Y	Y	Y
Sample adaptive offset (SAO): Improves encoded video quality.	Y	Y	Y	Y
B frame: Improves encoded quality.	Y	Y	Y	Y
Multiple reference frames : Capability to use different reference frames	Y	Y	Y	Y

- **Y**: Supported, **N**: Not supported

NVENC AV1 Hardware Capabilities

Table 3: NVENC AV1 Hardware Capabilities

Hardware Features	Ampere and TU GPUs	Ada GPUs	Blackwell GPUs	Jetson Thor
Main profile: Capability to encode YUV 420 8-bit and 10-bit content up to 8192 x 8192 content.	N	Y	Y	N

- **Y**: Supported, **N**: Not supported

Chapter 3. NVENC Licensing Policy

As far as NVENC hardware encoding is concerned, NVIDIA GPUs are classified into two categories: "qualified" and "non-qualified". On qualified GPUs, the number of concurrent encode sessions is limited by available system resources (encoder capacity, system memory, video memory etc.). On non-qualified GPUs, the number of concurrent encode sessions is limited to 12 per system. This limit of 12 concurrent sessions per system applies to the combined number of encoding sessions executed on all non-qualified cards present in the system.

For a complete list of qualified and non-qualified GPUs, refer to https://developer.nvidia.com/nvidia-video-codec-sdk.

For example, on a system with one Quadro RTX4000 card (which is a qualified GPU) and three GeForce cards (which are non-qualified GPUs), the application can run N simultaneous encode sessions on Quadro RTX4000 card (where N is defined by the encoder/memory/hardware limitations) and 12 sessions on all the GeForce cards combined. Thus, the limit on the number of simultaneous encode sessions for such a system is $N + 12$.

Chapter 4. NVENC Performance

With every generation of NVIDIA GPUs (Turing, Ampere, Ada, and Blackwell), NVENC performance has increased steadily. *NVENC H264 Performance* provides *indicative* NVENC performance on Turing, Ampere, Ada, Blackwell, and Jetson Thor GPUs for different presets and rate control modes (these two factors play a major role in determining the performance and quality). Note that performance numbers in *NVENC H264 Performance* for H264, *NVENC HEVC Performance* for HEVC, *NVENC AV1 Performance* for AV1 are measured on GeForce hardware with assumptions listed under the table. The performance varies across GPU classes (e.g. Quadro, Tesla), and scales (almost) linearly with the clock speeds for each hardware.

Some NVIDIA GPUs may have multiple NVENC engines per chip. This increases the aggregate encoder performance of the GPU. NVIDIA driver takes care of load balancing among multiple NVENC engines on the chip, so that applications don't require any special code to take advantage of multiple encoders and automatically benefit from higher encoder capacity on higher-end GPU hardware. The encode performance listed in *NVENC H264 Performance* for H264, *NVENC HEVC Performance* for HEVC, *NVENC AV1 Performance* for AV1 is given *per NVENC engine*. Thus, if the GPU has multiple NVENCs, multiply the corresponding number in *NVENC H264 Performance* for H264, *NVENC HEVC Performance* for HEVC, *NVENC AV1 Performance* for AV1 by the number of NVENCs per chip to get aggregate maximum performance (applicable only when running multiple simultaneous encode sessions). Note that unless Split Frame Encoding is enabled, performance with single encoding session cannot exceed performance per NVENC, regardless of the number of NVENCs present on the GPU. Multi NVENC Split Frame Encoding is a feature introduced in SDK12.0 on Ada GPUs for HEVC and AV1. Refer to the NVENC Video Encoder API Programming Guide for more details on this feature.

NVENC hardware natively supports multiple hardware encoding contexts with negligible context-switching penalty. As a result, subject to the hardware performance limit and available memory, an application can encode multiple videos simultaneously. NVENCODE API exposes several presets, rate control modes and other parameters for programming the hardware. A combination of these parameters enables video encoding at varying quality and performance levels. In general, one can trade performance for quality and vice versa.

NVENC H264 encoding performance in frames/second (fps)

Table 1: NVENC H264 Performance

Preset	RC Mode	Tuning Info	Turing	Ampere	Ada	Blackwell	Jetson Thor
P1	CBR	LL	855	868	910	977	724
P1	VBR	HQ	833	846	885	948	713

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Preset	RC Mode	Tuning Info	Turing	Ampere	Ada	Black-well	Jetson Thor
P3	CBR	LL	600	613	652	718	529
P3	VBR	HQ	602	617	647	708	527
P5	CBR	LL	271	273	291	323	236
P5	VBR	HQ	264	266	283	317	230
P7	CBR	LL	229	231	247	264	219
P7	VBR	HQ	207	213	211	227	202

NVENC HEVC encoding performance in frames/second (fps)

Table 2: NVENC HEVC Performance

Preset	RC Mode	Tuning Info	Turing	Ampere	Ada	Black-well	Jetson Thor
P1	CBR	LL	932	943	1055	1134	860
P1	VBR	HQ	920	939	1037	1119	850
P3	CBR	LL	463	467	494	529	402
P3	VBR	HQ	552	557	706	947	579
P5	CBR	LL	305	307	343	506	279
P5	VBR	HQ	334	335	411	521	336
P7	CBR	LL	306	308	343	464	279
P7	VBR	HQ	171	171	181	181	148

NVENC AV1 encoding performance in frames/second (fps)

Table 3: NVENC AV1 Performance

Preset	RC Mode	Tuning Info	Ada	Blackwell
P1	CBR	LL	1090	1076
P1	VBR	HQ	741	957
P3	CBR	LL	774	798
P3	VBR	HQ	549	678
P5	CBR	LL	512	624
P5	VBR	HQ	440	552

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Preset	RC Mode	Tuning Info	Ada	Blackwell
P7	CBR	LL	356	395
P7	VBR	HQ	323	401

- *Resolution/Input Format/Bit depth*: 1920 × 1080/YUV 4:2:0/8-bit
- Above measurements are made using the following GPUs: RTX 8000 for Turing, RTX 3090 for Ampere, RTX 4090 for Ada, RTX 5070 Ti for Blackwell, and NVIDIA® Jetson Thor™ for Thor GPUs. All measurements are done at the highest video clocks as reported by nvidia-smi (i.e. 1950 MHz, 1950 MHz, 2415 MHz, 2257 MHz for RTX 8000, RTX 3090, RTX 4090, and RTX 5070 Ti respectively). The performance should scale according to the video clocks as reported by nvidia-smi for other GPUs of every individual family. Information on nvidia-smi can be found at https://developer.nvidia.com/nvidia-system-management-interface.
- *Software*: Windows 11, Video Codec SDK v13.1; Thor GPU performance is measured on NVIDIA® Jetson™ Linux
- CBR: Constant bitrate rate control mode, VBR: Variable bitrate rate control mode, LL : Low latency tuning info, HQ: High quality tuning info

Chapter 5. Programming NVENC

The NVENCODE API provides access to the video encoding features of NVENC described in the previous chapters and provides control over encoding parameters. The NVENC hardware takes YUV/RGB as input and generates H.264/HEVC/AV1 compliant video bit streams.

Refer to the SDK release notes for information regarding the required driver version.

Refer to the documents and the sample applications included in the SDK package for details on how to program NVENC.

Chapter 6. FFmpeg Support

FFmpeg is the most popular multimedia transcoding tool used extensively for video and audio transcoding.

The video hardware accelerators in NVIDIA GPUs can be effectively used with FFmpeg to significantly speed up the video decoding, encoding and end-to-end transcoding at very high performance. For more information on how to use NVENC or NVDEC with FFmpeg, please refer to the FFmpeg guide in the Video Codec SDK.

Note that FFmpeg is open-source project and its usage is governed by specific licenses and terms and conditions for FFmpeg.

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